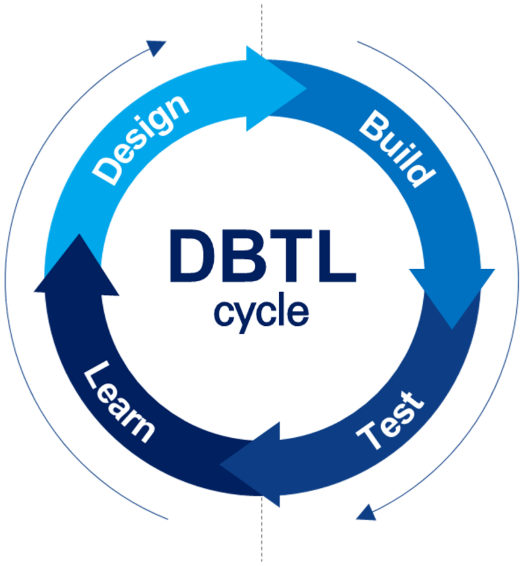
Engineering



Wet Lab

Our project follows the synthetic biology paradigm of iterative refinement. We decomposed the complex challenge of Small-Cell Lung Cancer (SCLC) therapy into four modular cycles, using Escherichia coli Nissle 1917 (EcN) as our core chassis.

As the project is currently only in the preliminary stage, this section only includes the Design and Build and Test(Plan) parts, as well as a Learn section identifying potential failure points based on our current engineering foresight.

Cycle 1: Chassis Selection & Adhesion Optimization

Design: We required a chassis capable of deep-lung colonization and efficient protein secretion. While *Lactobacillus plantarum* was considered for its immunomodulatory benefits, we selected Escherichia coli Nissle 1917 (EcN) for its superior genetic tractability and established Type I Secretion System (T1SS). To improve EcN’ s native lung retention, we designed a module to heterologously express the P30 adhesin from *Mycoplasma pneumoniae*.

Build: We designed **Plasmid B**, which incorporates the `LPP-OmpA-P30` fusion protein under the control of the `PJ23100` constitutive promoter.

Test: To validate the adhesion module, we have designed *in vitro* binding assays using human bronchial epithelial cells (e.g., BEAS-2B) to quantify the retention of EcN expressing the P30 adhesin after standardized washing. Additionally, the specificity of the active targeting module will be tested by comparing the binding affinity of DLL3-nanobody-functionalized bacteria against DLL3-positive SCLC cell lines versus DLL3-negative controls.

Learn (Potential Failures):

Insufficient Adhesion: If P30-mediated anchoring is too weak to resist mucociliary clearance, we will iterate by testing higher-affinity adhesins from other pulmonary pathogens.

Steric Hindrance: If the surface-displayed DLL3 nanobodies are shielded by the P30 protein, we will redesign the linkers to optimize the spatial orientation of the targeting ligands.

Cycle 2: The Lactate-Responsive Survival Switch

Design: To ensure environmental containment and tumor specificity, we leveraged the high lactate levels in the SCLC TME. Our design creates a survival switch by deleting the endogenous thyA gene, making the bacteria dependent on external thymidine or internal synthesis triggered by lactate.

Build: We designed a circuit where the essential thyA gene is driven by the lactate-responsive promoter PlldR. For laboratory expansion, we built a dual-control mechanism using the ParaC promoter to allow growth in the presence of arabinose *in vitro*.

Test: The functionality of the lactate-responsive survival circuit will be verified through growth curve analysis in media with varying lactate concentrations (0–20 mM) to simulate the tumor microenvironment versus healthy tissue. We will also perform escape simulations by transferring the bacteria to lactate-free and thymidine-free environments to measure the clearance rate, ensuring the PlldR-driven *thyA* expression provides robust biological containment.

Learn (Potential Failures):

Promoter Leakiness: If PlldR shows high basal expression in low-lactate healthy tissues, we will implement a Suicide Switch using a Toxin-Antitoxin system to enforce stricter death outside the TME.

Metabolic Load: If thyA expression imposes a significant metabolic burden, we will tune the Ribosome Binding Site (RBS) to lower protein production to the minimum viable threshold.

Cycle 3: Precision Therapeutic Secretion (FuncPept)

Design: We designed a smart peptide (FuncPept) that targets MYC protein aggregation. To prevent off-target toxicity, we designed a masking sequence that is only cleaved by Legumain, a protease overexpressed in SCLC. Additionally, we incorporated a Tuftsin (TKPR) sequence to facilitate macrophage-mediated clearance of amyloid fibers after therapy.

Build: We constructed the genetic sequence for the therapeutic payload, fusing the masked FuncPept-Tuftsin complex to the HlyA C-terminal signal sequence for export via the T1SS.

Test: Validation of the therapeutic module will begin with Western Blot analysis of the culture supernatant to confirm that the HlyA-tagged peptide is successfully exported via the Type I Secretion System (T1SS). To test the smart logic gate, the masked FuncPept will be incubated with recombinant Legumain at an acidic pH (4.0–6.0) to simulate the lysosomal environment, followed by mass spectrometry to verify precise cleavage at the AAN site. Finally, the kinetics of MYC protein aggregation induced by the released peptide will be monitored using Thioflavin T (ThT) fluorescence assays, while the recruitment efficiency of the Tuftsin sequence will be measured through macrophage phagocytosis assays.

Learn (Potential Failures):

Secretion Bottlenecks: If the T1SS machinery (HlyB/D) fails to export the complex peptide, we will test alternative secretion pathways or optimize the signal peptide sequence.

Activation Delay: If Legumain cleavage is too slow, we will use directed evolution to optimize the AAN site for higher protease affinity.

Incomplete Clearance: If Tuftsin fails to recruit sufficient macrophages, we will iterate by adding auxiliary chemoattractants to the payload.

Cycle 4: Dosage Optimization

Design: We propose a dynamic intervention strategy for SCLC treatment, using LDH level as a core monitoring indicator.The initial grading thresholds for LDH will be determined from published literature and standard clinical reference ranges.The intervention scheme includes drug dosage, administration frequency and timing, with serum LDH, CCL2 concentration, and macrophage polarization status as key readouts to evaluate therapeutic response.

Build: We will establish a preclinical intervention system for SCLC, and construct a graded intervention protocol based on LDH levels.LDH grading thresholds will be recalibrated using reference values from different laboratories and real-world clinical cohort data to improve reliability and applicability.Combination regimens will be constructed with defined component ratios, and preliminary safety and efficacy parameters will be set.

Test: We will perform quantitative measurements of LDH levels, serum CCL2 concentration, and macrophage polarization profiles before and after intervention.These dynamic changes will be used to assess the effectiveness of the current dosage, frequency and timing of administration.We will also evaluate pathological outcomes and safety indicators to determine the upper limit of intervention cycles and the maximum safe dose.For combination regimens, we will test different synergistic ratios and compare therapeutic and safety profiles.

Learn: By comparing predicted and observed responses, we will iteratively optimize dosage, frequency and timing of administration.The LDH grading system will be further refined to better match the clinical and pathological characteristics of SCLC.The maximum safe dose, intervention cycle limits, and optimal synergistic ratio of the combination regimen will be refined using preclinical efficacy and safety data.Lessons from this cycle will be used to update the grading criteria and administration protocol, forming a closed-loop, clinically adaptable intervention system.

Dry Lab

Cycle 1: Targeted Aerosolization Design

Design: Effective treatment of SCLC requires the bacteria to reach the bronchial and distal airway regions. Our design focused on a vibrating mesh nebulizer capable of generating aerosol particles with a specific aerodynamic diameter to ensure deep-lung deposition.

Build: We developed the specifications for a nebulizer system that allows for the adjustment of oscillation frequencies, which directly controls the particle size distribution of bacterial suspension.

Test:

> **Particle Size Characterization**: We have designed protocols to measure the Mass Median Aerodynamic Diameter (MMAD) and Geometric Standard Deviation (GSD) of the bacterial aerosol using a Laser Diffraction Particle Size Analyzer. The goal is to verify that the droplets fall within the 1–5 μm range required for distal airway deposition.
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> **Bacterial Viability Post-Nebulization**: We will collect the generated aerosol using a glass impinger and perform Colony Forming Unit (CFU) plating to determine the Survival Fraction of the bacteria after they are subjected to the mechanical vibration and shear stress of the mesh.

Learn (Potential Failures):

Suboptimal Deposition: If the MMAD is too large (leading to upper airway deposition) or too small (leading to exhalation of the bacteria), we will iterate by redesigning the mesh aperture size or adjusting the ultrasonic oscillation frequency of the nebulizer.

High Mechanical Mortality: If the CFU counts show significant bacterial death after nebulization, we will investigate the use of bio-friendly oscillation patterns with reduced power outputs or optimize the suspension buffer with osmoprotectants to stabilize the bacterial cell membranes.

Mesh Clogging: If the high concentration of bacteria or surface proteins leads to biofilm formation or clogging of the vibrating mesh, we will implement a self-cleaning pulse-wave cycle into the nebulizer's control software.